

IGNASHINO, Russian Federation

WMO: 306860

Lat: 53.467N

Lon: 122.400E

Elev: 295

StdP: 97.83

Time Zone: 9.00 (E09)

Period: 94-18

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-41.4	-39.3	-43.8	0.1	-41.1	-41.7	0.1	-39.1	7.7	-19.3	7.0	-20.0	0.4	0	0.624

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.1	30.9	20.0	28.8	19.3	26.7	18.7	21.7	27.6	20.7	26.1	19.8	24.7	1.8	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.9	15.2	24.2	19.0	14.3	23.0	18.1	13.5	22.3	65.3	27.8	61.6	26.1	58.3	24.6	27.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
7.4	6.2	5.1	-43.6	34.1	2.0	2.3	-45.0	35.7	-46.1	37.1	-47.2	38.3	-48.7	40.0		
			-43.5	23.9	1.8	1.6	-44.8	25.1	-45.8	26.0	-46.8	26.9	-48.1	28.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	-3.1	-28.1	-22.8	-12.2	0.6	9.3	16.7	19.5	16.9	8.9	-2.2	-17.9	-27.5
	DBStd	17.88	5.23	5.23	6.80	4.77	3.69	3.41	2.87	3.46	4.33	5.53	7.69	5.64
	HDD10.0	5582	1182	919	689	283	57	1	0	2	70	379	838	1162
	HDD18.3	7941	1440	1152	947	532	281	71	22	68	282	638	1088	1421
	CDD10.0	783	0	0	0	1	35	202	293	214	38	0	0	0
	CDD18.3	101	0	0	0	0	1	21	57	23	0	0	0	0
	CDH23.3	1424	0	0	0	1	36	459	638	280	10	0	0	0
	CDH26.7	450	0	0	0	0	6	149	221	74	0	0	0	0
	WSAvg	1.6	1.3	1.3	1.8	2.2	2.2	1.6	1.4	1.5	1.7	1.8	1.4	1.2
Precipitation	PrecAvg	421	3	4	7	20	33	64	104	101	49	19	11	6
	PrecMax	635	9	11	19	43	68	177	221	252	119	54	24	12
	PrecMin	274	0	0	1	1	4	16	37	34	6	1	2	1
	PrecStd	89	2	3	5	11	18	40	41	53	24	13	6	3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-11.8	-2.7	8.7	20.1	27.2	33.1	34.2	31.5	24.9	15.1	1.1	-9.3
		MCWB	-13.2	-5.5	2.6	9.3	14.5	20.6	20.8	20.8	16.5	8.1	-1.7	-10.2
	2%	DB	-14.3	-7.6	5.1	15.8	23.9	30.5	31.2	28.7	22.0	11.7	-1.7	-12.3
		MCWB	-15.3	-9.3	0.2	7.3	12.4	19.4	20.1	20.0	15.1	5.9	-3.4	-13.4
	5%	DB	-16.7	-10.5	2.2	12.7	21.4	28.1	29.3	26.6	19.5	8.9	-3.8	-15.4
		MCWB	-17.5	-11.9	-1.6	5.4	11.4	17.9	20.0	19.3	13.7	4.2	-5.1	-16.2
	10%	DB	-19.1	-13.0	-0.2	9.8	18.7	25.9	27.0	24.3	17.1	6.3	-6.3	-18.5
		MCWB	-19.6	-14.1	-3.1	3.6	10.3	17.0	19.5	18.1	11.8	2.6	-7.5	-19.0
	0.4%	WB	-13.2	-5.5	2.8	10.2	15.6	22.7	23.2	22.8	18.1	8.6	-1.4	-10.1
		MCDB	-12.1	-2.9	8.1	19.2	24.1	30.5	28.5	29.0	23.3	13.9	0.4	-9.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	-15.2	-9.1	0.4	7.7	13.8	20.6	22.0	21.2	16.1	6.5	-3.2	-13.4
		MCDB	-14.2	-7.6	4.6	15.3	21.0	27.7	27.8	26.4	20.3	10.9	-1.9	-12.3
	5%	WB	-17.4	-11.9	-1.4	5.7	12.4	19.3	21.1	20.1	14.7	4.7	-5.1	-16.2
		MCDB	-16.6	-10.5	1.9	11.8	19.3	26.1	26.5	24.7	17.8	8.2	-3.8	-15.4
	10%	WB	-19.6	-14.1	-3.1	4.1	11.1	18.0	20.3	19.0	13.0	2.8	-7.5	-19.0
		MCDB	-19.1	-12.9	-0.2	9.4	17.7	24.4	25.1	22.8	16.5	6.2	-6.3	-18.5
Mean Daily Temperature Range	5% DB	MDBR	12.7	17.5	18.5	14.5	15.2	15.5	13.1	12.9	13.9	13.0	12.8	11.1
		MCDBR	13.5	18.4	18.2	19.7	21.6	20.6	18.8	17.2	17.5	16.2	12.7	11.7
		MCWBR	12.6	16.5	13.7	11.8	11.4	10.0	8.6	8.8	10.6	10.8	10.8	10.9
	5% WB	MCDBR	13.4	18.2	17.3	18.0	18.0	17.4	13.9	14.1	14.5	15.8	12.5	11.7
		MCWBR	12.5	16.3	13.1	11.0	10.1	9.1	7.3	7.9	9.4	10.8	10.6	10.9
Clear-Sky Solar Irradiance	taub		0.256	0.287	0.336	0.466	0.536	0.489	0.462	0.419	0.366	0.334	0.290	0.264
	taud		2.130	2.093	2.033	1.856	1.803	2.040	2.175	2.280	2.364	2.302	2.149	2.057
	Ebn at Noon		707	798	829	753	721	768	785	800	803	741	645	595
	Edh at Noon		67	98	130	179	200	159	137	117	95	80	66	60
All-Sky Solar Radiation	RadAvg		1.25	2.40	4.09	5.24	5.25	6.02	5.24	4.59	3.49	2.28	1.35	0.94
	RadStd		0.05	0.06	0.16	0.36	0.38	0.52	0.46	0.46	0.29	0.12	0.06	0.05

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (6 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page